PROPER USE OF MANOMETER

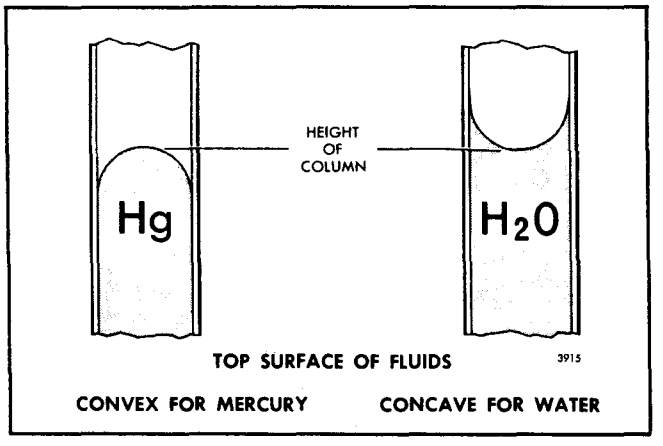

Fig. 4 - Comparison of Column Height for Mercury and Water Manometers

The U-tube manometer is a primary measuring device indicating pressure or vacuum by the difference in the height of two columns of fluid.

Connect the manometer to the source of pressure, vacuum, or differential pressure. When the pressure is imposed, add the number of inches one column of fluid travels up to the amount the other column travels down to obtain the pressure (or vacuum) reading.

PRESSURE CONVERSION CHART

1" water	=	.0735" mercury
1" water	=	.0361 psi
1" mercury	=	.491 psi
1" mercury	=	13.6" water
1 psi	=	27.7" water
1 psi	=	2.036" mercury

TABLE 7

The height of a column of mercury is read differently than that of a column of water. Mercury does not wet the inside surface; therefore, the top of the column has a convex meniscus (shape). Water wets the surface and therefore has a concave meniscus. A mercury column is read by sighting horizontally between the top of the convex mercury surface (Fig. 4) and the scale. A water manometer is read by sighting horizontally between the bottom of the concave water surface and the scale.

Should one column of fluid travel further than the other column, due to minor variations in the inside diameter of the tube or to the pressure imposed, the accuracy of the reading obtained is not impaired.

The manometer reading may be converted into other units of measurement by use of the pressure conversion Table 7.

ENGINE ELECTRICAL GENERATING SYSTEM

Whenever trouble is indicated in the engine electrical generating system, the following quick checks can be made to assist in localizing the cause:

1. A fully charged battery and low charging rate indicates normal alternator-regulator operation.

2. A low battery and high charging rate indicates normal alternator-regulator operation.

3. A fully charged battery and high charging rate condition usually indicates the voltage regulator is set too high or is not limiting the alternator output. A high charging rate to a fully charged battery will damage the battery and other electrical components.

4. A low battery and low or no charging rate condition could be caused by: Loose connections or damaged wiring, defective battery or alternator, or a defective regulator or improper regulator setting.

Contact an authorized *Detroit Diesel Allison Service Outlet* if more information is needed.